PREMIER LEAGUE GOALKEEPER ANALYTICS

Which Goalkeeper Performance Metrics Are Most Strongly Associated with Clean Sheets?

DATA ANALYTICS PORTFOLIO PROJECT – MICROSOFT EXCEL & DATA VISUALISATION

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SOURCE: FBREF PREMIER LEAGUE STATISTICS

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EXECUTIVE SUMMARY

This project investigates which publicly available goalkeeper statistics are most strongly associated with clean-sheet performance in the Premier League. The analysis was completed using Microsoft Excel and involved cleaning raw data, calculating standardised per-90 statistics, conducting correlation analysis and presenting the findings through a dashboard.

The purpose of the project was not simply to rank goalkeepers, but to explore which performance indicators showed the strongest statistical relationships with clean sheets. The analysis also considered the limitations of using clean sheets as a measure of individual goalkeeper ability, as team defensive quality, tactics and shot quality can significantly influence results.

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INTRODUCTION

Evaluating goalkeeper performance is more complex than comparing total saves or clean sheets. Statistics are influenced by several external factors, including the quality of the defense in front of them, the tactical style of the team and the number and difficulty of shots faced.

Clean sheets are commonly used to assess goalkeeper performance, but they represent a team outcome rather than a purely individual one. A goalkeeper playing behind a strong defensive line may achieve more clean sheets while facing fewer difficult shots than a goalkeeper playing for a weaker team.

This project uses publicly available Premier League goalkeeper data to examine whether commonly used performance metrics are statistically associated with clean-sheet performance. Microsoft Excel was used to clean the data, calculate derived statistics, conduct correlation analysis and produce a visual dashboard.

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RESEARCH QUESTION & OBJECTIVES

Which publicly available goalkeeper performance metrics are most strongly associated with clean sheets in the Premier League?

Objectives:

- Clean and organize publicly available goalkeeper data.
- Calculate comparable per-90 performance statistics.
- Measure relationships between selected metrics and clean sheet performance.
- Create and Excel dashboard to present findings.
- Evaluate the limitations of the analysis.

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DATASET

The dataset used in this project was obtained from FBref, a publicly available football statistics website. It contained Premier League goalkeeper performance data, including appearances, minutes played, goals conceded, shots on target faced, saves, save percentage, clean sheets and penalty statistics.

The original dataset was preserved in a raw-data sheet before any cleaning or calculations were performed. A separate clean-data sheet was then created for the analysis. To reduce the effect of very small sample sizes, only goalkeepers who played at least 900 minutes were included in the main analysis. This ensured that regular and semi-regular players were not directly compared with goalkeepers who had only made a small number of appearances.

Category	**Information**
Competition & Data Source	Premier League & FBref
Software	Microsoft Excel
Minimum playing time	900 Minutes
Unit of analysis	Individual goalkeepers
Main outcome variable	Clean sheets per 90

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DATA PREPARATION AND CLEANING

Before carrying out the analysis, the raw dataset was reviewed and cleaned to improve consistency and accuracy. Unnecessary technical columns were removed from the working dataset, while the original data were retained separately.

Player names, club names and numerical columns were checked to ensure that each goalkeeper occupied one row, and each variable occupied one column. Missing values, duplicate entries and formatting errors were also reviewed.

Additional per-90 variables were calculated to make comparisons fairer between players with different amounts of playing time.

The Steps:

- Removed unnecessary columns.
- Checked for missing values.
- Checked for duplicate players.
- Standardised numerical formatting.
- Filtered out goalkeepers with fewer than 900 minutes.
- Calculated clean sheets per 90.
- Calculated saves per 90.
- Calculated shots on target faced per 90.
- Calculated penalty save percentage.

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DERIVED METRICS

Key risks include potential cultural misalignment and system integration challenges. Mitigation strategies include regular leadership communication and phased technology implementation.

Metric	**Formula**
Clean sheets per 90	Clean sheets / 90s played
Saves per 90	Saves / 90s played
Shots on target faced per 90	Shots on target faced / 90s played
Penalty save percentage	Penalties saved / penalties faced

Per-90 statistics were used because total figures are strongly affected by playing time. Standardising the metrics made comparisons between goalkeepers more meaningful.

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METHODOLOGY

Pearson correlation analysis was used to measure the direction and strength of the relationship between goalkeeper performance metrics and clean sheets per 90.

Correlation coefficients range from −1 to +1. A positive value means that the two variables tend to increase together, while a negative value means that one variable tends to decrease as the other increases. Values closer to either −1 or +1 indicate stronger relationships, while values closer to zero indicate weaker relationships.

Correlation measures association rather than causation. A strong relationship does not prove that one metric directly causes an increase or decrease in clean sheets.

Interpretation Table:

Absolute Correlation	**Interpretation**
0.00–0.19	Very weak
0.20–0.39	Weak
0.40–0.59	Moderate

0.60–0.79	Strong
0.80–1.00	Very Strong

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RESULTS

Correlation Analysis — Clean Sheets per 90

Metric	Correlation with CS/90	Absolute Strength	Interpretation	Caution
Save percentage	0.54	0.54	Moderate	Association only; team quality and shot difficulty are not controlled.
Goals against per 90	-0.84	0.84	Strong	Association only; team quality and shot difficulty are not controlled.
Saves per 90	0.67	0.67	Moderate	Association only; team quality and shot difficulty are not controlled.
Shots on target faced per 90	0.66	0.66	Moderate	Association only; team quality and shot difficulty are not controlled.
Clean-sheet percentage	0.85	0.85	Strong	Association only; team quality and shot difficulty are not controlled.
Penalty save percentage	0.34	0.34	Weak	Association only; team quality and shot difficulty are not controlled.

Premier League Goalkeeper Analysis Dashboard

Goalkeepers analysed
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Strongest available metric
Clean-sheet percentage

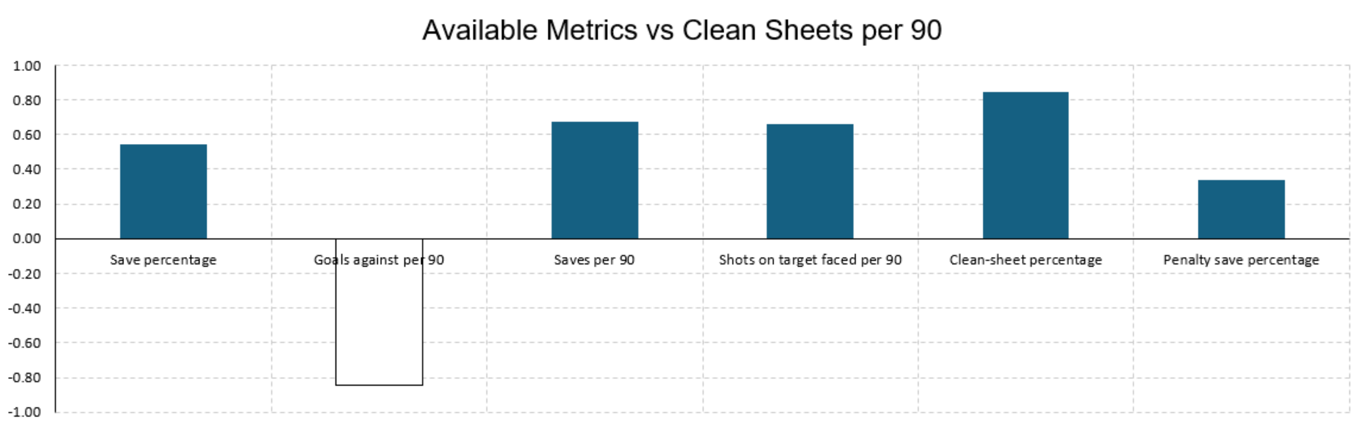
Highest absolute correlation
0.85

Headline finding

The strongest relationship in the available dataset is Clean-sheet percentage. This does not prove causation and should be tested with advanced metrics and team defensive controls.

Project limitations

- One-season sample
- Clean sheets depend on the whole team
- Uploaded file lacks PSxG and advanced sweeping data
- Correlation does not establish causation



Strongest relationships

The strongest relationships observed was between clean-sheet percentage and clean sheets per-90 along with goals against per 90 and clean sheets per-90. With correlation values of 0.85 and 0.84. This indicates a strong positive correlation.

Weakest relationship

The weakest relationship was between Penalty Save Percentage and clean sheets per 90, with a correlation coefficient of 0.34. This indicates little linear association between the two variables. A likely explanation is that penalty situations are relatively rare, meaning the metric is based on a small number of events for most goalkeepers.

Most interesting relationship

One particularly interesting finding was the relationship between save percentage and clean sheets per-90. Although this metric may be expected to have a strong correlation, the observed absolute strength had a correlation of 0.54 – moderate correlation.

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DISCUSSION

The results indicate that some goalkeeper metrics are more closely associated with clean sheets than others. However, the relationships must be interpreted carefully.

Save percentage may provide useful information about shot-stopping efficiency, but it does not account for shot difficulty. A goalkeeper facing mostly low-quality shots may record a stronger save percentage than one facing frequent high-quality chances.

Saves per 90 also require careful interpretation. A high number of saves may indicate strong individual performance, but it may also show that the goalkeeper plays behind a defense that allows a large number of shots.

Goals against per 90 is likely to be closely connected with clean-sheet performance because both measures involve goals conceded. This relationship may therefore be partly mechanical rather than providing an independent measure of goalkeeper quality.

Overall, the findings show why goalkeeper performance should be evaluated using multiple metrics rather than a single statistic.

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LIMITATIONS

First, clean sheets are influenced by the performance of the entire team. Defensive quality, possession, tactical structure and opposition strength can all affect the number of chances a goalkeeper faces.

Second, the project uses data from a single season. Relationships may change across different seasons, leagues or tactical environments.

Third, the dataset does not include advanced goalkeeper statistics such as post-shot expected goals prevented, cross-stopping percentage or defensive actions outside the penalty area.

Fourth, penalty save percentage is based on a small number of events for many goalkeepers, making the measure highly sensitive to sample size.

Finally, correlation analysis does not establish causation and does not control for other variables.

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FUTURE IMPPROVEMENTS

The analysis could be extended by using multiple seasons of data and including advanced goalkeeper metrics. Post-shot expected goals prevented would provide a stronger measure of shot-stopping performance because it considers the quality and placement of shots faced.

Team-level defensive variables could also be introduced to separate goalkeeper performance from the quality of the defence. Further analysis could use multiple regression to examine the relationship between several variables simultaneously.

The project could also be extended to compare goalkeeper performance across different leagues or to analyse changes in performance over time.

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CONCLUSION

This project investigated which publicly available goalkeeper metrics were most strongly associated with clean-sheet performance in the Premier League.

Using Microsoft Excel, the raw data were cleaned and standardised before per-90 statistics and Pearson correlations were calculated. The analysis found that [insert strongest genuine result].

However, the results should be interpreted as statistical associations rather than evidence of causation. Clean sheets are affected by goalkeeper ability, team defensive quality, tactics and the difficulty of shots faced.

The project demonstrates practical skills in data cleaning, quantitative analysis, data visualisation, dashboard development and professional reporting.

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REFERENCES

- FBref. Premier League Goalkeeper Statistics.
- Microsoft. Excel documentation and statistical functions.